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United States Patent	12397202
Kind Code	B2
Date of Patent	August 26, 2025
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Golf ball with increased core hardness gradient and reduced spin

Abstract

Compositions including a hardness agent, a water releasing agent, and a complexing agent and golf ball cores made from such compositions having a tailored hardness gradient are disclosed. The type and concentration of the components in the composition affect the hardness, hardness gradient, and compression of cores made from the composition and, thus, can be used to produce a golf ball having desirable performance characteristics.

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Appl. No.: 18/385568

Filed: October 31, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250135292 A1	May. 01, 2025

Publication Classification

Int. Cl.: A63B37/06 (20060101); A63B37/00 (20060101)

U.S. Cl.:

CPC A63B37/0063 (20130101); A63B37/0051 (20130101); A63B37/00621 (20200801); A63B2209/00 (20130101)

Field of Classification Search

CPC: A63B (37/0063); A63B (37/0051)

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Background/Summary

FIELD OF THE INVENTION

(1) The present disclosure relates generally to compositions for use in golf ball cores that provide for manipulation of the hardness gradient of such cores. More particularly, the present disclosure provides compositions and golf ball cores made from such compositions that provide an ability to tailor and/or improve certain ball performance results when such cores are used in a golf ball. In some respects, the present disclosure relates to golf ball cores with tailorable hardness gradients that, when used in golf balls, provide the ability to achieve one or more desired performance characteristics including, for example, spin on driver and short distance shots.

BACKGROUND OF THE INVENTION

(2) The performance of a golf ball is affected by a variety of factors including the materials, weight, size, dimple pattern, and external shape of the golf ball. As a result, golf ball manufacturers are constantly improving or tweaking the performance of golf balls by adjusting the materials and construction of the ball as well as the dimple pattern and dimple shape.

(3) For example, the resiliency and rebounding (Coefficient of Restitution) performance of the golf ball are generally driven by the composition and construction of the core. The spin rate and feel of the ball also are important properties that are affected by the composition and construction of the

core. The spin rate refers to the rate of rotation of the golf ball after being hit with a club. Two factors that affect the spin rate of a golf ball are the hardness gradient of the core, i.e., the difference in hardness between the geometric center and outer surface of the core, and the compression of the core, i.e., how much the core deflects under a given load. Generally, changing the compression of a golf ball or golf ball core alters the spin rate of the golf ball similarly for both driver shots and short distance shots. For example, increasing the compression of a golf ball or golf ball core may cause a 10 percent increase in spin rates for both short distance and driver shots. In contrast, changing the hardness gradient of the golf ball core may affect the spin rate differently for driver shots and short distance shots. More specifically, an increase in the hardness gradient may result in a 15 percent increase in spin rate on short distance shots but only a 5 percent increase in spin rate on driver shots.

(4) Most professionals and highly skilled amateurs (i.e., those who can control the spin of a golf ball) generally prefer balls with high spin rates to allow for better control in and around the green and draw and fade on approach shots. Indeed, these balls are beneficial for short distance shots made with irons and wedges. In contrast, recreational players who cannot necessarily control the spin of the ball will likely find that it is easier to play with a golf ball with low spin because the spin from a golf ball with a high spin rate can create more shot dispersion, i.e., more stray off to the left or right of the centerline, especially if the ball is hooked or sliced.

(5) Meanwhile, the “feel” of the ball generally refers to the sensation that a player experiences when striking the ball with the club. Most players prefer balls having a soft feel, because the players experience a more natural and comfortable sensation when the club face makes contact with these balls. Balls having a softer feel are particularly desirable when making short shots around the green, because the player senses more with such balls. The feel of the ball primarily depends upon the hardness and compression of the ball.

(6) Accordingly, there remains a need for golf ball cores having a hardness gradient that can be tailored to produce desired performance characteristics such as resiliency, rebounding, spin, and feel. In this aspect, it would be advantageous to tailor the core of a golf ball such that the finished golf ball has the desired amount of spin on short distance and driver shots for players having various levels of expertise as well as other advantageous properties, features, and benefits. For example, it would be beneficial to tailor the core of a golf ball so as to reduce shot dispersion for amateur players. Similarly, it would be advantageous to tailor the core of a golf ball such that the finished golf ball gives highly skilled players greater control over the shot. The present disclosure provides compositions for use in golf ball cores and golf balls containing such cores that allow for manipulation of the hardness gradient and, thus, manipulation or tailoring of desired performance characteristics.

SUMMARY OF THE INVENTION

(7) The problems expounded above, as well as others, are addressed by the following inventions, although it is to be understood that not every embodiment of the inventions described herein will address each of the problems described above.

(8) In some embodiments, the present disclosure provides a golf ball, including: a core formed from a composition including: a base rubber; a hardening agent, wherein the hardening agent includes tricyclic compound and a carboxylic acid, and wherein the tricyclic compound includes at least one carbon-carbon double bond; and a water releasing agent, and a cover layer disposed about the rubber layer.

In some aspects, the hardening agent may be present in an amount of about 0.5 and about 5 parts per hundred rubber. In other aspects, the hardening agent includes abietic acid, palustric acid, levopimaric acid, or combinations thereof. In still other aspects, the tricyclic compound includes a conjugated diene. In yet other aspects, the water releasing agent includes a metal salt hydrate. In some aspects, the water releasing agent is present in an amount of about 0.5 and about 5.0 parts per hundred rubber.

(9) The composition may include a complexing agent. In some aspects, the complexing agent includes a metal oxide. In other aspects, the metal oxide is zinc oxide. In still other aspects, the core includes a zinc rosinate complex. In yet other aspects, the core has a geometric center having a hardness H.sub.C, a geometric surface having a hardness H.sub.S, and a hardness gradient H.sub.Gr equal to the difference in the geometric center hardness and the surface hardness, wherein the hardness gradient is between about 20 Shore C and about 40 Shore C. The H.sub.C may be in the range of about 35 Shore C to about 65 Shore C, H.sub.S is in the range of about 60 Shore C to about 90 Shore C, and H.sub.S is greater than H.sub.C.

(10) In other embodiments, the present disclosure relates to a golf ball, including: a core formed from a rubber formulation including: a base rubber; a hardening agent including a tricyclic compound and a carboxylic acid, wherein the tricyclic compound includes a conjugated diene; a water releasing agent; and a complexing agent, wherein the complexing agent includes a metal oxide, wherein the core has a geometric center hardness H.sub.C, a surface hardness H.sub.S, and a hardness gradient H.sub.Gr equal to the difference between H.sub.C and H.sub.S and between about 20 and 45, and wherein the core includes a metal rosinate complex; and a cover layer disposed about the core.

In some aspects, the hardening agent includes abietic acid, palustric acid, levopimaric acid, or combinations thereof. In other aspects, the water releasing agent is metal salt hydrate. In still other aspects, the metal salt hydrate is present in an amount of about 0.5 and about 5.0 parts per hundred rubber and the hardening agent is present in an amount of about 0.5 to about 5 parts per hundred rubber.

(11) In yet other embodiments, the present disclosure relates to a golf ball, including: a core formed from a rubber formulation including: a base rubber; a hardening agent including a 20-carbon tricyclic carboxylic acid, isomers and diastereomers thereof, or combinations thereof; and a water releasing agent, wherein the hardening agent and water releasing agent are present in the rubber formulation in a ratio of about 1:1 to about 1:5, wherein the core has a geometric center hardness H.sub.C, a surface hardness H.sub.S, and a hardness gradient H.sub.Gr equal to the difference between H.sub.C and H.sub.S and between about 10 and 50; and a cover layer disposed about the core.

In some aspects, the formulation further includes a complexing agent. In other aspects, the complexing agent includes a metal oxide. In yet other aspects, the core includes a metal rosinate complex of Formula I:

(12)
$$HA \cdot X$$
 (I)

where HA is a deprotonated form of the hardening agent and X is the metal in the complexing agent. In still other aspects, X includes zinc, iron, aluminum, titanium, magnesium, or zirconium.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further features and advantages of the invention can be ascertained from the following detailed description that is provided in connection with the drawings described below:

(2) FIG. 1 is a cross-sectional view of a two-piece golf ball in accordance with an embodiment of the present disclosure;

(3) FIG. 2 is a cross-sectional view of a three-piece golf ball in accordance with an embodiment of the present disclosure;

(4) FIG. 3 is a cross-sectional view of a four-piece golf ball in accordance with an embodiment of the present disclosure;

(5) FIG. 4 is a cross-sectional view of a five-piece golf ball in accordance with an embodiment of the present disclosure; and

(6) FIGS. 5-7 are graphical interpretations of properties of cores made in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(7) The present disclosure relates to compositions that may be used to produce a tailorable hardness gradient, cores including such compositions that possess a desired hardness gradient, and golf balls including such cores. In some respects, the tailorable nature of the hardness gradient in cores made in accordance with the present disclosure provides the ability to increase or reduce driver spin when compared to a conventional golf ball hit under the same conditions. In addition, the tailorable hardness gradient in the core may be used to provide greater control on long shots as well as approach shots and greenside play.

(8) While the golf ball core is functionally different from the other layers of the golf ball and operates somewhat independently, the core of the present disclosure greatly influences the overall performance of the finished golf ball including such a core. Without being bound by any particular theory, since a core typically represents about 90 percent of the golf ball weight, performance characteristics of a finished golf ball that contains the core of the present disclosure may be tailored by changing the core composition. For example, altering the core composition and, thus, the hardness gradient, may have a significant effect on long shots, e.g., shots off of a driver, and approach shots, e.g., shots made with irons and wedges. In fact, adjusting the hardness gradient of cores made in accordance with this present disclosure, even in seemingly minor amounts, can significantly affect how a golf ball performs on long and short distance shots. Similarly, adjusting the hardness gradient of the core may allow for tailoring of other properties of the finished golf ball. The core formulations, cores, golf balls, and resulting performance characteristics are discussed in greater detail below.

(9) Core Formulations

(10) The present disclosure provides golf balls having single- or multi-layered cores made from core rubber formulations that result in increased “positive” hardness gradients across the core. In some embodiments, the core rubber formulations of the present disclosure include a base rubber, a hardening agent, a water releasing agent, a cross-linking agent, and a free radical initiator. The core rubber formulations may also optionally include additives, such as one or more of a metal oxide, metal fatty acid or fatty acid, antioxidant, soft and fast agent, or fillers. Concentrations of components are in parts per hundred (phr) unless otherwise indicated. As used herein, the term, “parts per hundred,” also known as “phr” or “pph” is defined as the number of parts by weight of a particular component present in a mixture, relative to 100 parts by weight of the polymer component. Mathematically, this can be expressed as the weight of an ingredient divided by the total weight of the polymer, multiplied by a factor of 100.

(11) Base Rubber

(12) The core rubber formulations of the present disclosure include a base rubber. In some embodiments, the base rubber may include natural and synthetic rubbers and combinations of two or more thereof. Examples of natural and synthetic rubbers suitable for use as the base rubber include, but are not limited to, polybutadiene, polyisoprene, ethylene propylene rubber (EPR), ethylene-propylene-diene (EPDM) rubber, grafted EPDM rubber, styrene-butadiene rubber, styrenic block copolymer rubbers (such as “SI”, “SIS”, “SB”, “SBS”, “SIBS”, and the like, where “S” is styrene, “I” is isobutylene, and “B” is butadiene), polyalkenamers such as, for example, polyoctenamer, butyl rubber, halobutyl rubber, polystyrene elastomers, polyethylene elastomers, polyurethane elastomers, polyurea elastomers, metallocene-catalyzed elastomers and plastomers, copolymers of isobutylene and p-alkylstyrene, halogenated copolymers of isobutylene and p-alkylstyrene, copolymers of butadiene with acrylonitrile, polychloroprene, alkyl acrylate rubber, chlorinated isoprene rubber, acrylonitrile chlorinated isoprene rubber, and combinations of two or more thereof.

(13) For example, the core may be formed from a rubber formulation that includes polybutadiene

as the base rubber. Polybutadiene is a homopolymer of 1,3-butadiene. The double bonds in the 1,3-butadiene monomer are attacked by catalysts to grow the polymer chain and form a polybutadiene polymer having a desired molecular weight. Any suitable catalyst may be used to synthesize the polybutadiene rubber depending upon the desired properties. In one embodiment, a transition metal complex (for example, neodymium, nickel, or cobalt) or an alkyl metal such as alkyl lithium is used as a catalyst. Other catalysts include, but are not limited to, aluminum, boron, lithium, titanium, and combinations thereof. The catalysts produce polybutadiene rubbers having different chemical structures. In a cis-bond configuration, the main internal polymer chain of the polybutadiene appears on the same side of the carbon-carbon double bond contained in the polybutadiene. In a trans-bond configuration, the main internal polymer chain is on opposite sides of the internal carbon-carbon double bond in the polybutadiene. The polybutadiene rubber can have various combinations of cis- and trans-bond structures. For example, the polybutadiene rubber may have a 1,4 cis-bond content of at least 40 percent. In another embodiment, the polybutadiene rubber has a 1,4 cis-bond content of greater than 80 percent. In still another embodiment, the polybutadiene rubber has a 1,4 cis-bond content of greater than 90 percent. In general, polybutadiene rubbers having a high 1,4 cis-bond content have high tensile strength and rebound.

(14) The polybutadiene rubber may have a relatively high or low Mooney viscosity. Generally, polybutadiene rubbers of higher molecular weight and higher Mooney viscosity have better resiliency than polybutadiene rubbers of lower molecular weight and lower Mooney viscosity. However, as the Mooney viscosity increases, the milling and processing of the polybutadiene rubber generally becomes more difficult. Blends of high and low Mooney viscosity polybutadiene rubbers may be prepared as is described in U.S. Pat. Nos. 6,982,301 and 6,774,187, the disclosures of which are hereby incorporated by reference, and used in accordance with the present disclosure. In general, the lower limit of Mooney viscosity may be about 30 or 35 or 40 or 45 or 50 or 55 or 60 or 70 or 75 and the upper limit may be about 80 or 85 or 90 or 95 or 100 or 105 or 110 or 115 or 120 or 125 or 130. For example, the polybutadiene used in the rubber formulation may have a Mooney viscosity of about 30 to about 80 or about 40 to about 60.

(15) Examples of commercially available polybutadiene rubbers that can be used in rubber formulations in accordance with the present disclosure, include, but are not limited to, BR 01 and BR 1220, available from BST Elastomers of Bangkok, Thailand; SE BR 1220LA and SE BR1203, available from DOW Chemical Co of Midland, Mich.; BUDENE 1207, 1207s, 1208, and 1280 available from Goodyear, Inc of Akron, Ohio; BR 01, 51 and 730, available from Japan Synthetic Rubber (JSR) of Tokyo, Japan; BUNA CB 21, CB 22, CB 23, CB 24, CB 25, CB 29 MES, CB 60, CB Nd 60, CB 55 NF, CB 70 B, CB KA 8967, and CB 1221, available from Lanxess Corp. of Pittsburgh, Pa.; BR1208, available from LG Chemical of Seoul, South Korea; UBEPOL BR130B, BR150, BR150B, BR150L, BR230, BR360L, BR710, and VCR617, available from UBE Industries, Ltd. of Tokyo, Japan; EUROPRENE NEOCIS BR 60, INTENE 60 AF and P30AF, and EUROPRENE BR HV80, available from Polimeri Europa of Rome, Italy; KBR 01, NdBr 40, NdBR-45, NdBr 60, KBR 710S, KBR 710H, and KBR 750, available from Kumho Petrochemical Co., Ltd. Of Seoul, South Korea; DIENE 55NF, 70AC, and 320 AC, available from Firestone Polymers of Akron, Ohio; and PBR-Nd Group II and Group III, available from Nizhnekamskneftekhim, Inc. of Nizhnekamsk, Tartarstan Republic.

(16) In another embodiment, the core is formed from a rubber formulation including butyl rubber. Butyl rubber is an elastomeric copolymer of isobutylene and isoprene. Butyl rubber is an amorphous, non-polar polymer with good oxidative and thermal stability, good permanent flexibility, and high moisture and gas resistance. Generally, butyl rubber includes copolymers of about 70 percent to about 99.5 percent by weight of an isoolefin, which has about 4 to 7 carbon atoms, for example, isobutylene, and about 0.5 percent to about 30 percent by weight of a conjugated multiolefin, which has about 4 to 14 carbon atoms, for example, isoprene. The resulting copolymer contains about 85 percent to about 99.8 percent by weight of combined isoolefin and

about 0.2 percent to about 15 percent of combined multiolefin. A commercially available butyl rubber suitable for use in rubber formulations in accordance with the present disclosure includes Bayer Butyl 301 manufactured by Bayer AG.

(17) The rubber formulations may include a combination of two or more of the above-described rubbers as the base rubber. In some embodiments, the rubber formulation of the present disclosure includes a blend of different polybutadiene rubbers. In this embodiment, the rubber formulation may include a blend of a first polybutadiene rubber and a second polybutadiene rubber in a ratio of about 5:95 to about 95:5. For example, the rubber formulation may include a first polybutadiene rubber and a second polybutadiene rubber in a ratio of about 10:90 to about 90:10 or about 15:85 to about 85:15 or about 20:80 to about 80:20 or about 30:70 to about 70:30 or about 40:60 to about 60:40. In other embodiments, the rubber formulation may include a blend of more than two polybutadiene rubbers or a blend of polybutadiene rubber(s) with any of the other elastomers discussed above.

(18) In other embodiments, the rubber formulation used to form the core includes a blend of polybutadiene and butyl rubber. In this embodiment, the rubber formulation may include a blend of polybutadiene and butyl rubber in a ratio of about 10:90 to about 90:10. For example, the rubber formulation may include a blend of polybutadiene and butyl rubber in a ratio of about 10:90 to about 90:10 or about 20:80 to about 80:20 or about 30:70 to about 70:30 or about 40:60 to about 60:40. In other embodiments, the rubber formulation may include polybutadiene and/or butyl rubber in a blend with any of the other elastomers discussed above.

(19) In further embodiments, the rubber formulation used to form the core includes a blend of polybutadiene and EPDM rubber or grafted EPDM rubber as the base rubber. In still further embodiments, the rubber formulations may include a combination of polybutadiene rubber and EPDM rubber as the base rubber. In this embodiment, the EPDM may be included in the rubber formulation in an amount of about 0.1 to about 20 or about 1 to about 15 or about 3 to about 10 parts by weight per 100 parts of the total rubber. For example, EPDM may be included in the rubber formulation in an amount of about 5 parts by weight per 100 parts of the total rubber. In still further embodiments, the core formulations may combine EPDM rubber and two or more different types of polybutadiene rubber, such as two or more different types of high cis-1,4 polybutadiene, as the base rubber.

(20) The rubber formulations include the base rubber in an amount of 100 phr. That is, when more than one rubber component is used in the rubber formulation as the base rubber, the sum of the amounts of each rubber component should total 100 phr. In some embodiments, the rubber formulations include polybutadiene rubber as the base rubber in an amount of 100 phr. In other embodiments, the rubber formulations include polybutadiene rubber and a second rubber component. In this embodiment, the polybutadiene rubber may be used in an amount of about 80 to about 99.9 parts by weight per 100 parts of the total rubber and the second rubber component may be used in an amount of about 0.1 to about 20 parts by weight per 100 parts of the total rubber. In further embodiments, the polybutadiene rubber may be used in an amount of about 85 to about 99 parts by weight per 100 parts of the total rubber and the second rubber component may be used in an amount of about 1 to about 15 parts by weight per 100 parts of the total rubber. In yet other embodiments, the polybutadiene rubber may be used in an amount of about 90 to about 97 parts by weight per 100 parts of the total rubber and the second rubber component may be used in an amount of about 3 to about 10 parts by weight per 100 parts of the total rubber. In still further embodiments, the polybutadiene rubber may be used in an amount of about 94 to about 96 parts by weight per 100 parts of the total rubber and the second rubber component may be used in an amount of about 4 to about 6 parts by weight per 100 parts of the total rubber. In some embodiments, the second rubber component is EPDM rubber.

(21) The base rubber may be used in the rubber formulation in an amount of at least about 5 percent by weight based on total weight of the rubber formulation. In some embodiments, the base

rubber is included in the rubber formulation in an amount within a range having a lower limit of about 10 percent or 20 percent or 30 percent or 40 percent or 50 percent or 55 percent and an upper limit of about 60 percent or 70 percent or 80 percent or 90 percent or 95 percent or 100 percent. For example, the base rubber may be present in the rubber formulation in an amount of about 30 percent to about 80 percent by weight based on the total weight of the rubber formulation. In another example, the rubber formulation includes about 40 percent to about 70 percent base rubber based on the total weight of the rubber formulation.

(22) Hardening Agent

(23) The rubber formulations of the present disclosure include one or more hardening agents. Without being bound to any particular theory, the hardening agent may affect the hardness of the core and the hardness gradient across the core.

(24) The hardening agent is an aliphatic or aromatic organic compound including a tricyclic compound and a carboxylic acid. In some embodiments, the hardening agent includes a 20-carbon tricyclic carboxylic acid, an isomer thereof (including, but not limited to, a double bond isomer), a diastereoisomer thereof, and combinations thereof. Without being bound by any particular theory, different isomers and/or additional functionalization of the hardening agent may affect the hardness of the core differently and produce different hardness gradients across the core. In this aspect, the tricyclic compound may include a conjugated diene. In other aspects the tricyclic compound includes at least one carbon-carbon double bond. In still other aspects, the tricyclic compound includes at least two carbon-carbon double bonds. In yet other aspects, the tricyclic compound includes three carbon-carbon double bonds.

(25) The molecular weight of the hardening agent may range from about 285 to about 325 g/mol. In one embodiment, the molecular weight of the hardening agent ranges from 290 to about 310 g/mol. In another embodiment, the molecular weight of the hardening agent ranges from 295 to about 305 g/mol.

(26) Suitable hardening agents include abietic acid, palustric acid, levopimaric acid, neoabietic acid, pimaric acid, isopimaric acid, sandaracopimaric acid, dihydroabietic acid, isomers and diastereomers thereof, and combinations thereof. In some embodiments, the hardening agent in the composition includes one or more of the following compounds:

(27) ##STR00001##

(28) In other embodiments, the hardening agent in the composition includes one or more of the following compounds:

(29) ##STR00002##

(30) In still other embodiments, the hardening agent is the following compound:

(31) ##STR00003##

(32) The hardening agent may be included in the rubber formulation in varying amounts depending on the desired characteristics of the golf ball core. For example, the hardening agent may be used in an amount of 0.01 to about 15 parts by weight per 100 parts of the total rubber. In some embodiments, the hardening agent may be used in an amount of 0.01 to about 10 parts by weight per 100 parts of the total rubber. In other embodiments, the hardening agent may be used in an amount of 0.01 to about 5 parts by weight per 100 parts of the total rubber. In one embodiment, the rubber formulation of the core includes about 0.05 to about 4 or about 0.1 to about 3.5 or about 0.5 to 3.4 parts by weight hardening agent per 100 parts of the total rubber. In another embodiment, the hardening agent is included in the rubber formulation in an amount of about 1 to about 3 parts by weight per 100 parts of the total rubber. In still another embodiment, the rubber formulation includes about 0.5 to about 3.5 or 0.7 to about 3.3 or about 0.8 to about 3.2 or about 0.9 to about 3.1 parts by weight hardening agent per 100 parts of the total rubber.

(33) Water Releasing Agent

(34) The core rubber formulations of the present disclosure include a water releasing agent. A “water releasing agent,” as used herein, refers to a compound having at least one water molecule

available for release during the curing process. When the free radical initiator decomposes to generate decomposition heat at the time of curing of the core, the temperature near the surface of the core is kept substantially the same as the temperature of the mold, while the temperature near the center of the core increases because of the accumulated decomposition heat of the free radical initiator. Without being bound by any particular theory, it is believed that, by adding a water releasing agent to the core rubber formulation that can release water at the desired curing temperature, the water can promote further decomposition of the free radical initiator and deactivation of radicals at the center of the core, which, in turn, results in a difference in crosslinking density and an increased hardness gradient between the center and the surface.

(35) The water releasing agent of the present disclosure has a moisture content capable of releasing a sufficient amount of water to promote decomposition of the free radical initiator and deactivation of radicals during the curing process. The moisture content in the water releasing agent can be calculated as the mass of water in the water releasing agent divided by the total weight of the water releasing agent. In some embodiments, the water releasing agent has a moisture content (in its molecular form) of at least about 5 percent by mass. In further embodiments, the water releasing agent has a moisture content ranging from about 5 percent by mass to about 95 percent by mass. In still further embodiments, the water releasing agent has a moisture content ranging from about 10 percent by mass to about 90 percent by mass. In yet further embodiments, the water releasing agent has a moisture content ranging from about 15 percent by mass to about 85 percent by mass. In further embodiments, the water releasing agent has a moisture content of at least about 50 percent by mass. For example, the water releasing agent has a moisture content of about 50 percent by mass to about 95 percent by mass.

(36) In some embodiments, the water releasing agent of the present disclosure may be a metal sulfate hydrate having one or more waters of hydration capable of being released during the reactions of the present disclosure. In one embodiment, the metal may be an alkaline earth metal. For example, the metal may be calcium, magnesium, beryllium, strontium, barium, or radium. In one embodiment, the metal of the metal sulfate hydrate is calcium. In another embodiment, the metal of the metal sulfate hydrate is magnesium. In further embodiments, the metal may be a transition metal or a post-transition metal. For instance, the metal may be zinc, copper, iron, cobalt, manganese, chromium, nickel, aluminum, zirconium, cadmium, indium, or vanadium. In still further embodiments, the metal may be neodymium or lanthanum.

(37) The metal sulfate hydrate may have any number of waters of hydration. In some embodiments, the metal sulfate hydrate may have from 0.5 to ten waters of hydration. For instance, the metal sulfate hydrate may be a hemihydrate, monohydrate, dihydrate, trihydrate, tetrahydrate, pentahydrate, hexahydrate, heptahydrate, octahydrate, nonahydrate, or decahydrate. In further embodiments, the metal sulfate hydrate may have from one to seven waters of hydration. In still further embodiments, the metal sulfate hydrate may have from one to four waters of hydration. In yet further embodiments, the metal sulfate hydrate may have from one to three waters of hydration. In other embodiments, the metal sulfate hydrate may have two waters of hydration. For example, in one embodiment, the metal sulfate hydrate may be a dihydrate. In still further embodiments, the metal sulfate hydrate may be a heptahydrate (i.e., having seven waters of hydration).

(38) Examples of suitable metal sulfate hydrates contemplated for use as the water releasing agent in accordance with the present disclosure include, but are not limited to, calcium sulfate hemihydrate ($\text{CaSO}_4 \cdot 0.5\text{H}_2\text{O}$), calcium sulfate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), magnesium sulfate heptahydrate ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$), zinc sulfate dihydrate ($\text{ZnSO}_4 \cdot 2\text{H}_2\text{O}$), zinc sulfate heptahydrate ($\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$), vanadium oxide sulfate hydrate ($\text{VO}_2 \cdot \text{SO}_4 \cdot x\text{H}_2\text{O}$), neodymium sulfate hydrate ($\text{Nd}_2(\text{SO}_4)_3 \cdot x\text{H}_2\text{O}$), lanthanum oxalate hydrate ($\text{La}_2(\text{C}_2\text{O}_4)_3 \cdot x\text{H}_2\text{O}$), zinc sulfate monohydrate ($\text{ZnSO}_4 \cdot \text{H}_2\text{O}$), zirconium sulfate hydrate ($\text{Zr}(\text{SO}_4)_2 \cdot x\text{H}_2\text{O}$),

beryllium sulfate tetrahydrate ($\text{BeSO}_4 \cdot 4\text{H}_2\text{O}$), manganese sulfate hydrate ($\text{MnSO}_4 \cdot 4\text{H}_2\text{O}$), iron sulfate hydrate ($\text{FeSO}_4 \cdot 4\text{H}_2\text{O}$), cobalt sulfate hydrate ($\text{CoSO}_4 \cdot 4\text{H}_2\text{O}$), cadmium sulfate monohydrate ($\text{CdSO}_4 \cdot \text{H}_2\text{O}$), cadmium sulfate octahydrate ($\text{CdSO}_4 \cdot 8\text{H}_2\text{O}$), indium sulfate nonahydrate ($\text{InSO}_4 \cdot 9\text{H}_2\text{O}$), nickel sulfate heptahydrate ($\text{NiSO}_4 \cdot 7\text{H}_2\text{O}$), nickel sulfate hexahydrate ($\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$), aluminum sulfate hydrate ($\text{Al}_2(\text{SO}_4)_3 \cdot x\text{H}_2\text{O}$), and copper sulfate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$). In some embodiments, the water releasing agent is a metal sulfate dihydrate such as calcium sulfate dihydrate and zinc sulfate dihydrate.

(39) The water releasing agent may be included in the rubber formulation in varying amounts depending on the desired characteristics of the golf ball core. For example, the water releasing agent may be used in an amount of 0.1 to about 10 or about 0.5 to about 8 or about 0.8 to about 5 parts by weight water releasing agent per 100 parts of the total rubber. In one embodiment, the rubber formulation of the core includes about 0.8 to about 4 parts by weight water releasing agent per 100 parts of the total rubber. In some embodiments, the water releasing agent is included in the rubber formulation in an amount of about 0.3 to about 6 or about 0.5 to about 3.5 or about 0.7 to about 4 or about 0.5 to about 5 parts by weight water releasing agent per 100 parts of the total rubber. For example, in some embodiments, the water releasing agent is present in the core rubber formulation in an amount of about 1 phr to about 3 phr.

(40) In some respects, the amount of water releasing agent in the rubber formulation may differ based on the compound used as the water releasing agent. In particular, the amount of water releasing agent included may depend on the moisture content and moisture availability in the water releasing agent. When the water releasing agent has a greater moisture content, less water releasing agent may be needed in the rubber formulation to give the golf ball core the desired performance characteristics. For example, more water releasing agent may be required when the water releasing agent is calcium sulfate dihydrate, which has a moisture content of approximately 21 percent, than if water releasing agent is magnesium sulfate heptahydrate, which has a moisture content of about 51 percent. The moisture availability refers to how readily the water molecules dissociate from the water releasing agent. If the water releasing agent used has a high moisture availability, less water releasing agent may be needed than if the water releasing agent has a low moisture availability. And, as discussed above, the amount of water releasing agent in the rubber formulation may differ based on the amount, type, or both of the hardening agent.

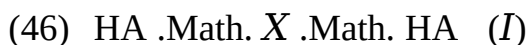
(41) In some embodiments, it may be desirable to formulate the rubber formulation of the golf ball core based on the water content of the rubber formulation instead of or in addition to the concentration of the water releasing agent. In this aspect, the water content of the rubber formulation can be determined by multiplying the concentration of water releasing agent in parts by weight per 100 parts of the total rubber by the moisture content of the water releasing agent. For example, if the water releasing agent is included in the rubber formulation at about 2 parts by weight per 100 parts of the total rubber and the moisture content in the water releasing agent is 50 percent, the water content of the rubber formulation is 1 part by weight per 100 parts of the total rubber. As such, the desired or targeted water content of the rubber formulation may be used to determine the concentration (and/or type of) water releasing agent.

(42) In some embodiments, the rubber formulation has a water content of about 0.01 to about 10 or about 0.05 to about 7 or about 0.1 to about 5 parts by weight water per 100 parts of the total rubber. In other embodiments, the water content of the rubber formulation is about 0.1 to about 2.5 or about 0.2 to about 2 or about 0.5 to about 1.5 parts by weight per 100 parts of the total rubber. In other embodiments, the water content of the rubber formulation is about 0.05 to about 1 or about 0.1 to about 0.5 or about 0.2 to about 0.6 parts by weight per 100 parts of the total rubber. In further embodiments, the rubber formulation has a water content of about 1 to about 3 or about 1.5 to about 2.5 or about 1 to about 2 parts by weight per 100 parts of the total rubber.

(43) In addition, the amount of water releasing agent used in the rubber formulation may affect the amount of hardening agent used in the rubber formulation. More specifically, when included in the rubber formulation, the water releasing agent itself may affect the hardness gradient and, thus, may allow less hardening agent to be employed. For example, when the water releasing agent is included in an amount of about 1 to about 5 parts by weight, the hardening agent may be included in an amount of about 0.1 to about 3.5 parts by weight per 100 parts of the total rubber. In the alternative, when the water releasing agent is included in an amount of about 0.1 to about 3 parts by weight, the hardening agent may be included in an amount of about 0.3 to about 2.5 parts by weight per 100 parts of the total rubber.

(44) Complexing Agent

(45) The composition of the present disclosure also includes a complexing agent. In some aspects, the complexing agent may be used to form a metal rosinate complex according to the general Formula I below:



where HA is the deprotonated form of the hardening agents discussed above (or conjugate base of the hardening agent) and X is the metal in the complexing agent. In this aspect, the complexing agent for use with the present disclosure may include a metal oxide (XO). Suitable metal oxides include, but are not limited to, zinc oxide, iron oxide, aluminum oxide, titanium oxide, magnesium oxide, and zirconium oxide. As such, X in Formula I may be zinc, iron, aluminum, titanium, magnesium, or zirconium.

(47) The complexing agent may be present in the composition in an amount of about 0.5 to about 10 parts by weight per 100 parts of the total rubber. In one embodiment, the rubber formulation includes the complexing agent in an amount of about 1 to about 15 or about 2 to about 10 parts by weight per 100 parts of the total rubber. In another embodiment, the rubber formulation includes the complexing agent in an amount of about 3 to about 8 or about 4 to about 6 parts by weight per 100 parts of the total rubber. In yet another embodiment, the rubber formulation includes the complexing agent in an amount of about 2 to about 7.

(48) In some embodiments, the complexing agent includes zinc oxide. Without being bound to any particular theory, the use of zinc oxide in combination with the hardening agent in the compositions of the present disclosure allow a zinc rosinate complex to form. In this aspect, the formed core would include zinc rosinate. The content of zinc in the complex may vary depending on the amount of complexing agent used, but in some aspects ranges from about 1 percent to about 10 percent by weight based on the total weight of the complex. For example, the zinc rosinate complex (Formula I where X=zinc) may include about 2 percent to about 9 percent by weight zinc based on the total weight of the complex.

(49) Crosslinking Co-Agent

(50) The rubber formulations further include a reactive cross-linking co-agent. Suitable co-agents include, but are not limited to, metal salts of unsaturated carboxylic acids having from 3 to 8 carbon atoms; unsaturated vinyl compounds and polyfunctional monomers (e.g., trimethylolpropane trimethacrylate); phenylene bismaleimide; and combinations thereof. In one embodiment, the co-agent is one or more metal salts of acrylates, diacrylates, methacrylates, and dimethacrylates, wherein the metal is selected from magnesium, calcium, zinc, aluminum, lithium, and nickel. In another embodiment, the co-agent includes one or more zinc salts of acrylates, diacrylates, methacrylates, and dimethacrylates. For example, the co-agent may be zinc diacrylate (ZDA). In another embodiment, the co-agent may be zinc dimethacrylate (ZDMA). An example of a commercially available zinc diacrylate includes Dymalink® 526 manufactured by Cray Valley.

(51) The co-agent may be included in the rubber formulation in varying amounts depending on the desired characteristics of the golf ball core. For example, the co-agent may be used in an amount of about 5 to about 50 or about 10 to about 45 or about 15 to about 40 parts by weight per 100 parts of

the total rubber. In one embodiment, the rubber formulation of the core includes about 20 to about 45 parts by weight co-agent per 100 parts of the total rubber. In another embodiment, the rubber formulation includes about 25 to about 40 or about 28 to about 38 parts by weight co-agent per 100 parts of total rubber. In another embodiment, the co-agent is included in the rubber formulation of the core in an amount of about 30 to about 35 or about 27 to about 36 parts by weight per 100 parts of the total rubber. In still another embodiment, the rubber formulation includes about 25 to about 34 or about 26 to about 33 parts by weight co-agent per 100 parts of the total rubber.

(52) In some respects, the amount of co-agent in the rubber formulation may be altered based on the class of compounds, and the particular isomer within a class of compounds, used as the hardening agent. Without being bound to any particular theory, the concentration of co-agent may be altered to achieve the desired compression of the golf ball core when different hardening agents are used.

(53) Free Radical Initiator

(54) The core formulations may include a free radical initiator selected from an organic peroxide, a high energy radiation source capable of generating free radicals, or a combination thereof. Suitable organic peroxides include, but are not limited to, dicumyl peroxide; n-butyl-4,4-di(t-butylperoxy) valerate; 1,1-di(t-butylperoxy)3,3,5-trimethylcyclohexane; 2,5-dimethyl-2,5-di(t-butylperoxy) hexane; di-t-butyl peroxide; di-t-amyl peroxide; t-butyl peroxide; t-butyl cumyl peroxide; 2,5-dimethyl-2,5-di(t-butylperoxy)hexyne-3; di(2-t-butyl-peroxyisopropyl)benzene; dilauroyl peroxide; dibenzoyl peroxide; t-butyl hydroperoxide; and combinations thereof. In a particular embodiment, the free radical initiator is dicumyl peroxide, including, but not limited to Perkadox® BD-FF, commercially available from Akzo Nobel. In other embodiments, the free radical initiator is dimethyl tertbutyl peroxide, including, but not limited to Trigonox® 101-50D-PD, commercially available from Nouryon.

(55) Free radical initiators may be present in the rubber formulation in an amount of at least 0.05 parts by weight per 100 parts of the total rubber, or an amount within the range having a lower limit of 0.05 parts or 0.1 parts or 1 part or 1.25 parts or 1.5 parts or 2.5 parts or 5 parts by weight per 100 parts of the total rubber, and an upper limit of 2.5 parts or 3 parts or 5 parts or 6 parts or 10 parts or 15 parts by weight per 100 parts of the total rubber. For example, the rubber formulation may include peroxide free radical initiators in an amount of about 0.1 to about 10 or about 0.5 to about 6 or about 1 to about 5 parts by weight per 100 parts of the total rubber. In another example, the rubber formulation may include peroxide free radical initiators in an amount of about 0.3 to about 2 or about 0.4 to about 1.8 or about 0.5 to about 1.5 or about 0.6 to about 1.4 parts by weight per 100 parts of the total rubber. In yet another example, the rubber formulation may include peroxide free radical initiators in an amount of about 0.3 to about 0.9 parts by weight per 100 parts of the total rubber.

(56) Additives

(57) Radical scavengers such as a halogenated organosulfur, organic disulfide, or inorganic disulfide compounds may also be added to the rubber formulation. In one embodiment, a halogenated organosulfur compound included in the rubber formulation includes, but is not limited to, pentachlorothiophenol (PCTP) and salts of PCTP such as zinc pentachlorothiophenol (ZnPCTP). In another embodiment, ditolyl disulfide, diphenyl disulfide, dixylyl disulfide, 2-nitroresorcinol, and combinations thereof are added to the rubber formulation. An example of a commercially available radical scavenger includes Rhenogran® Zn-PTCP-72 manufactured by Rheine Chemie. The radical scavenger may be included in the rubber formulation in an amount of about 0.1 to about 1 part by weight per 100 parts of the total rubber. In one embodiment, the rubber formulation may include about 0.2 to about 0.8 parts by weight radical scavenger per 100 parts of the total rubber. In another embodiment, the rubber formulation may include about 0.3 to about 0.7 parts by weight radical scavenger per 100 parts of the total rubber.

(58) The rubber formulation may also include filler(s). Suitable non-limiting examples of fillers

include carbon black, clay and nanoclay particles, talc, glass (e.g., glass flake, milled glass, and microglass), mica and mica-based pigments (e.g., Iriodin® pearl luster pigments from The Merck Group), and combinations thereof. Metal oxide and metal sulfate fillers are also contemplated for inclusion in the rubber formulation. Suitable metal fillers include, for example, particulate, powders, flakes, and fibers of copper, steel, brass, tungsten, titanium, aluminum, magnesium, molybdenum, cobalt, nickel, iron, lead, tin, zinc, barium, bismuth, bronze, silver, gold, and platinum, and alloys and combinations thereof. Any of the metal oxides discussed above for use as the complexing agent may also be suitable for use as filler. Suitable metal sulfate fillers include, for example, barium sulfate and strontium sulfate.

(59) When included, the fillers may be in an amount of about 1 to about 25 parts by weight per 100 parts of the total rubber. In one embodiment, the rubber formulation includes at least one filler in an amount of about 5 to about 20 or about 8 to about 15 parts by weight per 100 parts of the total rubber. In another embodiment, the rubber formulation includes at least one filler in an amount of about 8 to about 14 or about 10 to about 16 parts by weight per 100 parts of the total rubber. In yet another embodiment, the rubber formulation includes at least one filler in an amount of about 10 to about 17 parts by weight per 100 parts of the total rubber. An example of a commercially available barium sulfate filler includes PolyWate® 325 manufactured by Cimbar Performance Minerals.

(60) In some embodiments, more than one type of filler may be included in the rubber formulation. For example, the rubber formulation may include a first filler in an amount from about 5 to about 20 or about 8 to about 17 parts by weight per 100 parts total rubber and a second filler in an amount from about 1 to about 10 or about 3 to about 7 parts by weight per 100 parts total rubber. In another example, the rubber formulation may include a first filler in an amount from about 7 to about 13 or about 9 to about 12 parts by weight per 100 parts total rubber and a second filler in an amount from about 2 to about 8 or about 4 to about 6 parts by weight per 100 parts total rubber. In yet another example, the rubber formulation may include a first filler in an amount from about 10 to about 15 or about 13 to about 14 parts by weight per 100 parts total rubber and a second filler in an amount from about 2 to about 9 or about 3 to about 7 parts by weight per 100 parts total rubber. In a further example, the rubber formulation may include a first filler in an amount from about 10 to about 15 or about 13 to about 14 parts by weight per 100 parts total rubber and a second filler in an amount from about 13 to about 18 or about 14 to about 16 parts by weight per 100 parts total rubber.

(61) Antioxidants, processing aids, accelerators (for example, tetra methylthiuram), dyes and pigments, wetting agents, surfactants, plasticizers, coloring agents, fluorescent agents, chemical blowing and foaming agents, defoaming agents, stabilizers, softening agents, impact modifiers, antiozonants, as well as other additives known in the art, may also be added to the rubber formulation. Examples of suitable processing aids include, but are not limited to, high molecular weight organic acids and salts thereof. Suitable organic acids are aliphatic organic acids, aromatic organic acids, saturated mono-functional organic acids, unsaturated monofunctional organic acids, multi-unsaturated mono-functional organic acids, and dimerized derivatives thereof. In one embodiment, the organic acids include, but are not limited to, caproic acid, caprylic acid, capric acid, lauric acid, stearic acid, behenic acid, erucic acid, oleic acid, linoleic acid, myristic acid, benzoic acid, palmitic acid, phenylacetic acid, naphthalenoic acid, and dimerized derivatives thereof. The salts of organic acids include the salts of barium, lithium, sodium, zinc, bismuth, chromium, cobalt, copper, potassium, strontium, titanium, tungsten, magnesium, cesium, iron, nickel, silver, aluminum, tin, or calcium, salts of fatty acids, particularly stearic, behenic, erucic, oleic, linoelic or dimerized derivatives thereof.

(62) Curing the Core Formulation

(63) The base rubber, hardening agent, water releasing agent, cross-linking agent, free radical initiator, fillers, and any other materials used in forming the core, in accordance with the present disclosure, may be combined to form a mixture by any type of mixing known to one of ordinary skill in the art. Suitable types of mixing include single pass and multi-pass mixing, and the like. A

single pass mixing process where ingredients are added sequentially is preferred, as this type of mixing tends to increase efficiency and reduce costs for the process. In embodiments where a free-radical initiator is used, it may be desirable to combine the hardening agent into the rubber formulation prior to adding the free-radical initiator.

(64) The rubber formulation may be cured using conventional curing processes. Non-limiting examples of curing processes suitable for use in accordance with the present disclosure include peroxide-curing, sulfur-curing, high-energy radiation, and combinations thereof.

(65) Golf Ball Properties

(66) Hardness

(67) The hardness of the geometric center of the core may be obtained according to the following: the core is first gently pressed into a hemispherical holder having an internal diameter approximately slightly smaller than the diameter of the core, such that the core is held in place in the hemispherical portion of the holder while concurrently leaving the geometric central plane of the center exposed. The core is secured in the holder by friction, such that it will not move during the cutting and grinding steps, but the friction is not so excessive that distortion of the natural shape of the core would result. The core is secured such that the parting line of the center is roughly parallel to the top of the holder. The diameter of the center is measured 90 degrees to this orientation prior to securing. A measurement is also made from the bottom of the holder to the top of the core to provide a reference point for future calculations. A rough cut is made slightly above the exposed geometric center of the core using a band saw or other appropriate cutting tool, making sure that the core does not move in the holder during this step. The remainder of the core, still in the holder, is secured to the base plate of a surface grinding machine. The exposed 'rough' surface is ground to a smooth, flat surface, revealing the geometric center of the core, which can be verified by measuring the height from the bottom of the holder to the exposed surface of the core, making sure that exactly half of the original height of the core, as measured above, has been removed to within 0.004 inches. Leaving the core in the holder, the geometric center of the core is confirmed with a center square and carefully marked, and the hardness is measured at the center mark according to ASTM D-2240.

(68) Additional hardness measurements at any distance from the geometric center of the core can then be made by drawing a line radially outward from the geometric center mark and measuring the hardness at any given distance along the line, typically in 2 mm increments from the center of the core. The hardness at a particular distance from the geometric center should be measured along at least two, preferably four, radial arms located 180° apart, or 90° apart, respectively, and then averaged. All hardness measurements performed on a plane passing through the geometric center are performed while the core is still in the holder and without having disturbed its orientation, such that the test surface is constantly parallel to the bottom of the holder, and thus also parallel to the properly aligned foot of the durometer.

(69) The outer surface hardness of the core (or any golf ball layer) is measured on the actual outer surface of the layer and is obtained from the average of a number of measurements taken from opposing hemispheres, taking care to avoid making measurements on the parting line of the core or on surface defects, such as holes or protrusions and preferably making the measurements prior to surrounding the layer of interest with an additional layer. Hardness measurements are made pursuant to ASTM D-2240 "Indentation Hardness of Rubber and Plastic by Means of a Durometer." Because of the curved surface, care must be taken to ensure that the golf ball or golf ball sub-assembly is centered under the durometer indenter before a surface hardness reading is obtained. A calibrated, digital durometer, capable of reading to 0.1 hardness units is used for the hardness measurements. The digital durometer must be attached to, and its foot made parallel to, the base of an automatic stand. The weight on the durometer and attack rate conforms to ASTM D-2240. It is worthwhile to note that, once an additional layer surrounds a layer of interest, the hardness of the layer of interest can be difficult to determine. Therefore, for purposes of the present

disclosure, when the hardness of a layer is needed after the inner layer has been surrounded with another layer, the test procedure for measuring a point located 1 mm from an interface is used.

(70) It should also be noted that there is a fundamental difference between “material hardness” and “hardness as measured directly on a golf ball” (or, as used herein, “surface hardness”). For purposes of the present disclosure, material hardness is measured according to ASTM D2240 and generally involves measuring the hardness of a flat “slab” or “button” formed of the material. Surface hardness as measured directly on a golf ball (or other spherical surface) typically results in a different hardness value. The difference in “surface hardness” and “material hardness” values is due to several factors including, but not limited to, ball construction (that is, core type, number of layers, and the like); ball (or ball sub-assembly) diameter; and the material composition of adjacent layers. It also should be understood that the two measurement techniques are not linearly related and, therefore, one hardness value cannot easily be correlated to the other. Shore hardness (for example, Shore C or Shore D hardness) was measured according to the test method ASTM D-2240.

(71) A golf ball core made from the rubber formulation of the present disclosure may have a hardness at the geometric center of the core, referred to herein as H.sub.C, that ranges from about 30 to about 80 Shore C. In one embodiment, the core has a hardness at its geometric center of about 35 to about 65 Shore C or about 40 to about 60 Shore C or about 45 to about 55 Shore C. In another embodiment, the core has a hardness at its geometric center of about 40 to about 55 Shore C or about 30 to about 60 Shore C.

(72) The hardness at the surface of the core, referred to herein as H.sub.S, may range from about 60 to about 100 Shore C. In one embodiment, the hardness at the surface of the core is about 70 to about 95 Shore C or about 72 to about 93 Shore C or about 75 to about 90 Shore C or about 80 to about 90 Shore C. In another embodiment, the hardness at the surface of the core is about 78 to about 92 Shore C.

(73) The direction of the hardness gradient is defined by the difference in hardness measurements taken at the geometric center and outer surfaces of the core. The geometric center hardness is readily determined according to the test procedures provided above. For example, the hardness of the outer surface of the core is also readily determined according to the procedures given herein for measuring the outer surface hardness of a golf ball layer, if the measurement is made prior to surrounding the core with additional layers.

(74) While the hardness gradient across the core will vary based on several factors including, but not limited to, the dimensions and formulations of the components, the core of the present disclosure has a “positive” hardness gradient (that is, the geometric center is softer than the outer surface of the core). More particularly, the term, “positive hardness gradient” as used herein means a hardness gradient of positive about 2 Shore C or greater, about 4 Shore C or greater, about 6 Shore C or greater, about 8 Shore C or greater, or about 10 Shore C or greater. In general, the hardness gradient may be determined by subtracting the hardness value of one component being measured (for example, the geometric center of the core, H.sub.C) from the hardness value of another component being measured (for example, the outer surface of the core, H.sub.S).

(75) The core of the present disclosure has a positive hardness gradient. In one embodiment, the core has a positive hardness gradient from the geometric center to the surface of the core of about 10 Shore C to 45 Shore C. In this aspect, the positive hardness gradient of the core is about 15 Shore C to about 45 Shore C. The rubber formulation of the core may be tailored to produce a desired hardness gradient in the core. In some embodiments, the positive hardness gradient of the core is about 20 to about 40 Shore C or about 22 Shore C to 38 Shore C or about 25 Shore C to about 35 Shore C.

(76) The hardness of the core may not increase linearly from the center of the core to the outer surface of the core. For example, one or more regions within the core may have a “zero” hardness gradient, i.e., the hardness values across the region are substantially the same. The term, “zero hardness gradient” as used herein means a hardness gradient of -2 Shore C to 2 Shore C,

preferably between about -1 Shore C and about 1 Shore C and may have a value of zero. In some embodiments, one or more regions of the core may also have a “negative” hardness gradient, i.e., the hardness values across the region may decrease from the inner edge of the region to the outer edge of the region.

(77) For example, the core, or a layer of the core if the core has multiple layers, may be characterized by three regions: an inner region, an intermediate region, and an outer region. Each of the inner region, intermediate region, and outer region may have its own hardness gradient. For a single-layer core, the inner region is the region of the core surrounding the center of the core and is characterized by positive hardness gradient of about 2 Shore C to about 25 Shore C. In some embodiments, the positive hardness gradient of the inner region of the core is about 6 Shore C to about 25 Shore C or about 16 Shore C to about 23 Shore C. In other embodiments, the positive hardness gradient of the inner region of the core is about 1 Shore C to about 13 Shore C or about 6 Shore C to about 11 Shore C. In further embodiments, the positive hardness gradient of the inner region of the core is about 5 Shore C to about 9 Shore C or about 6 Shore C to about 8 Shore C.

(78) The outer region of the core is the region of the core adjacent the surface of the core and may be characterized by a zero or positive hardness gradient from about -2 Shore C to about 28 Shore C. In some embodiments, the outer region may have a positive hardness gradient from 2 Shore C to about 27 Shore C or about 16 Shore C to about 27 Shore C or about 17 Shore C to about 22 Shore C. In other embodiments, the outer region may have a zero or positive hardness gradient from -2 Shore C to about 16 Shore C or about 2 Shore C to about 6 Shore C or about 10 Shore C to about 15 Shore C. In further embodiments, the outer region may have a zero or positive hardness gradient from -2 Shore C to about 14 Shore C or about 1 Shore C to about 8 Shore C or about 2 Shore C to about 6 Shore C.

(79) The intermediate region of the core is the region of the core between the inner region and the outer region and may be characterized by a negative, zero, or positive hardness gradient from about -10 to 8 Shore C. In some embodiments, the intermediate region may have a negative, zero, or positive hardness gradient from -7 to about 6 Shore C or about -6 to about 1 Shore C. In other embodiments, the intermediate region may have a positive hardness gradient from -7 to about 4 Shore C or about -2 to about 4 Shore C. In further embodiments, the intermediate region may have a negative or zero hardness gradient from -10 to about 0 Shore C or about -4 Shore C to about 0 Shore C.

(80) In some embodiments, a point or plurality of points measured along a “positive” gradient may be above or below a line fit through the gradient and its outermost and innermost hardness values. In an alternative embodiment, the hardest point along a particular steep “positive” gradient may be higher than the value at the innermost portion of the center (the geometric center) or outer surface of the core—as long as the outermost point (i.e., the outer surface of the core) is greater than the innermost point (i.e., the geometric center of the core), such that the “positive” gradients remain intact.

(81) Compression

(82) Several different methods can be used to measure compression, including Atti compression, Riehle compression, load/deflection measurements at a variety of fixed loads and offsets, and effective modulus (see, e.g., Compression by Any Other Name, Science and Golf IV, Proceedings of the World Scientific Congress of Golf (Eric Thain ed., Routledge, 2002) (J. Dalton). For purposes of the present disclosure, compression values are provided as measured by the Dynamic Compression Machine (“DCM”). The DCM applies a load to a ball component or a ball and measures the number of inches the core or ball is deflected at measured loads. A crude load/deflection curve is generated that is fit to the Atti compression scale that results in a number being generated that represents an Atti compression. The DCM does this via a load cell attached to the bottom of a hydraulic cylinder that is triggered pneumatically at a fixed rate (typically about 1.0 ft/s) towards a stationary core. Attached to the cylinder is an LVDT that measures the distance the

cylinder travels during the testing timeframe. A software-based logarithmic algorithm ensures that measurements are not taken until at least five successive increases in load are detected during the initial phase of the test.

(83) The SCDI is a slight variation of the DCM set up that allows determination of the pounds required to deflect a component or ball 10 percent of its diameter. With the SCDI, the goal is to obtain the pounds of force required to deflect a component or ball a certain number of inches. That amount of deflection is 10 percent of the component or ball diameter. The DCM is triggered, the cylinder deflects the component or ball by 10 percent of its diameter, and the DCM reports back the pounds of force required (as measured from the attached load cell) to deflect the component or ball by that amount. The SCDI value obtained is a single number in units of pounds.

(84) The compression of a core made from the rubber formulation of the present disclosure may range from about 20 to about 85 DCM. For example, the core compression may be about 30 to about 75 DCM or about 35 to about 85 DCM or about 35 to about 75 DCM. In another example, the core compression may range from about 40 to about 75 DCM or about 35 to about 80 DCM. In some embodiments, it may be desirable for a core comprising the rubber formulation of the present disclosure to have a compression from about 40 to about 70 DCM regardless of the hardening agent used.

(85) Diameter

(86) The diameter of the core may vary. In some embodiments, the core diameter may range from about 1.35 to about 1.65 inches. For example, the core may have a diameter of 1.39 to 1.62 inches. In embodiments where the core comprises two or more layers, the diameter of the inner layer of the core may range from about 0.25 to about 1.51 inches or from about 0.5 to about 1.2 inches or from about 0.6 to about 1.4 inches.

(87) Core Component Relationships

(88) In one embodiment, the amount of hardening agent and water releasing agent present in the rubber formulation and the number of double bonds in the tricyclic compound of the hardening agent are related to the hardness gradient of the core, according to the relationship shown in Equation I below:

$$(89) \frac{WRA_C * HA_C * HA_{DB}}{1 - \frac{1}{H_{Gr}}} \geq 1.8 \quad (I)$$

where WRA.sub.C represents the concentration of water releasing agent in the rubber formulation in parts per hundred; HA.sub.C represents the concentration of hardening agent in the rubber formulation in parts per hundred; H.sub.Gr represents the hardness gradient (Shore C) of the core or H.sub.S–H.sub.C, and H.sub.Gr≥20; and HA.sub.DB represents the number of carbon-carbon double bonds in the tricyclic compound of the hardening agent. For example, if the hardening agent is abietic acid, HA.sub.DB is equal to 2, and if the hardening agent is pimaric acid, HA.sub.DB is equal to 1.

(90) In another embodiment,

$$(91) 2 < \frac{WRA_C * HA_C * HA_{DB}}{1 - \frac{1}{H_{Gr}}} < 30$$

In still another embodiment

$$(92) 2 < \frac{WRA_C * HA_C * HA_{DB}}{1 - \frac{1}{H_{Gr}}} < 28$$

(93) In another aspect, the number of carbon-carbon double bonds in the tricyclic compound of the hardening agent present in the rubber formulation used to form the core is related to the hardness gradient (Shore C) of the core and ratio of the amount of the hardening agent to the complexing agent according to the relationship shown in Equation II below:

$$(94) \frac{HA_{DB} * R_{HA-CA}}{1 - \frac{1}{H_{Gr}}} > 0.2 \quad (II)$$

where H.sub.Gr represents the hardness gradient (Shore C) of the core or H.sub.S–H.sub.C, and H.sub.Gr≥20; HA.sub.DB represents the number of carbon-carbon double bonds in the hardening

agent; and R.sub.HA-CA represents the ratio of the amount of hardening agent to the amount of complexing agent in the rubber composition. In another embodiment,

$$(95) 0.2 < \frac{R_{HA-DB} * R_{HA-CA}}{1 - \frac{1}{H_{Gr}}} < 1.85$$

In still another embodiment,

$$(96) 0.4 < \frac{R_{HA-DB} * R_{HA-CA}}{1 - \frac{1}{H_{Gr}}} < 1.3$$

In yet another embodiment,

$$(97) 0.2 < \frac{R_{HA-DB} * R_{HA-CA}}{1 - \frac{1}{H_{Gr}}} < 0.6$$

(98) In some embodiments, the ratio of the amounts of hardening agent to water releasing agent and complexing agent present in the rubber formulation used to form the core are related to the hardness gradient of the core, according to the relationship shown in Equation IV below:

$$(99) 0.05 \leq \frac{R_{HA-WRA} * R_{HA-CA}}{1 - \frac{1}{H_{Gr}}} \leq 0.7 \quad (III)$$

where R.sub.HA-WRA represents the ratio of the amount of hardening agent to the amount of water releasing agent in the rubber composition; R.sub.HA-CA represents the ratio of the amount of hardening agent to the amount of complexing agent in the rubber composition; and H.sub.Gr represents the hardness gradient (Shore C) of the core or the difference between the hardness at the surface of the core and the hardness at the geometric center of the core (H.sub.S-H.sub.C), and $20 \leq H_{Gr} \leq 40$. In another embodiment,

$$(100) 0.06 \leq \frac{R_{HA-WRA} * R_{HA-CA}}{1 - \frac{1}{H_{Gr}}} \leq 0.65$$

(101) In other embodiments, the amounts of hardening agent and complexing agent to the amount of water releasing agent present in the rubber formulation used to form the core are in accordance with the relationship shown in Equation IV below:

$$(102) 2 \leq \frac{HA_C * CA_C}{WRA_C} \leq 8 \quad (IV)$$

where HA.sub.C represents the concentration of hardening agent in the rubber formulation in parts per hundred; CA.sub.C represents the concentration of complexing agent in the rubber formulation in parts per hundred; and WRA.sub.C represents the concentration of water releasing agent in the rubber formulation in parts per hundred.

(103) In yet another embodiment,

$$(104) 2 \leq \frac{HA_C * CA_C}{WRA_C} \leq 6$$

Golf Ball Construction

(105) Golf balls having various constructions may be made in accordance with the present disclosure. For example, golf balls having one-piece, two-piece, three-piece, four-piece, and five or more-piece constructions with the term “piece” refer to any core, cover, or intermediate layer of a golf ball construction. Representative illustrations of such golf ball constructions are provided and discussed further below. The term, “layer” as used herein means generally any spherical portion of the golf ball.

(106) In one embodiment, a golf ball of the present disclosure is a one-piece ball where the core and cover form a single integral layer. In another version, shown in FIG. 1, a golf ball of the present disclosure is a two-piece ball **10** comprising a single core layer **12** and a single cover layer **14**. As shown in FIG. 2, in one embodiment, the golf ball **20** comprises a core layer **22**, an intermediate layer **24**, and a cover layer **26**. In FIG. 2, the intermediate layer **24** can be considered an outer core layer, an inner cover layer, a mantle or casing layer, or any other layer disposed between the core **22** and the cover layer **26**. Referring to FIG. 3, in another embodiment, a four-piece golf ball **30** comprises an inner core layer **32**, an outer core layer **34**, an intermediate layer **36**, and an outer cover layer **38**. In FIG. 3, the intermediate layer **36** may be considered a casing or mantle layer, or inner cover layer, or any other layer disposed between the outer core layer **34** and the outer cover of the ball **38**. Referring to FIG. 4, in another version, a five-piece golf ball **40** comprises a three-layered core having an inner core layer **42**, an intermediate core layer **44**, an

outer core layer **46**, an inner cover layer **48**, and an outer cover layer **50**. As exemplified herein, a golf ball in accordance with the present disclosure can comprise any combination of any number of core layers, intermediate layers, and cover layers.

(107) The rubber formulations discussed above are suitable for use in the core or one or more of the core layers if multiple core layers are present. It is also contemplated that the rubber formulations disclosed herein may be used to form one or more of the layers of any of the one, two, three, four, or five, or more-piece (layered) balls described above. That is, any of the core layers, intermediate layers, and/or cover layers may comprise the rubber formulation of this disclosure. The rubber formulations of different layers may be the same or different. The diameter and thickness of the different layers along with properties such as hardness and compression may vary depending upon the construction and desired playing performance properties of the golf ball.

(108) Golf balls made in accordance with the present disclosure can be of any size, although the USGA requires that golf balls used in competition have a diameter of at least 1.68 inches. For play outside of United States Golf Association (USGA) rules, the golf balls can be of a smaller size. In one embodiment, golf balls made in accordance with the present disclosure have a diameter in the range of about 1.68 to about 1.80 inches.

(109) In contrast to the core, the cover of a golf ball plays less of a role on shots off of a driver. However, because the cover plays a significant role in generating spin on iron and wedge shots, the cover material and properties are still important. In this aspect, varied materials may be used in the construction of the intermediate and cover layers of golf balls according to the present disclosure. For example, a variety of materials may be used for forming the outer cover including, for example, polyurethanes; polyureas; copolymers, blends and hybrids of polyurethane and polyurea; olefin-based copolymer ionomer resins; polyethylene, including, for example, low density polyethylene, linear low density polyethylene, and high density polyethylene; polypropylene; rubber-toughened olefin polymers; acid copolymers, for example, poly(meth)acrylic acid, which do not become part of an ionomeric copolymer; plastomers; flexomers; styrene/butadiene/styrene block copolymers; styrene/ethylene-butylene/styrene block copolymers; dynamically vulcanized elastomers; copolymers of ethylene and vinyl acetates; copolymers of ethylene and methyl acrylates; polyvinyl chloride resins; polyamides, poly(amide-ester) elastomers, and graft copolymers of ionomer; cross-linked trans-polyisoprene and blends thereof; polyester-based thermoplastic elastomers; polyurethane-based thermoplastic elastomers; synthetic or natural vulcanized rubber; and combinations thereof.

(110) In one embodiment, the cover is formed from a polyurethane, polyurea, or hybrid of polyurethane-polyurea. When used as cover layer materials, polyurethanes and polyureas can be thermoset or thermoplastic. Thermoset materials can be formed into golf ball layers by conventional casting or reaction injection molding techniques. Thermoplastic materials can be formed into golf ball layers by conventional compression or injection molding techniques.

(111) Conventional and non-conventional materials may be used for forming intermediate layers of the ball including, for instance, ionomer resins, highly neutralized polymers, polybutadiene, butyl rubber, and other rubber-based core formulations, and the like. In one embodiment, the inner cover layer, i.e., the layer disposed between the core and the outer cover, includes an ionomer. In this aspect, ionomers suitable for use in accordance with the present disclosure may include partially neutralized ionomers and highly neutralized ionomers (HNPs), including ionomers formed from blends of two or more partially-neutralized ionomers, blends of two or more highly-neutralized ionomers, and blends of one or more partially-neutralized ionomers with one or more highly-neutralized ionomers. For purposes of the present disclosure, "HNP" refers to an acid copolymer after at least 70 percent of all acid groups present in the composition are neutralized.

(112) Preferred ionomers are salts of O/X- and O/X/Y-type acid copolymers, wherein O is an α -olefin, X is a C.sub.3-C.sub.8 α , β -ethylenically unsaturated carboxylic acid, and Y is a softening monomer. O is preferably selected from ethylene and propylene. X is preferably selected from

methacrylic acid, acrylic acid, ethacrylic acid, crotonic acid, and itaconic acid. Methacrylic acid and acrylic acid are particularly preferred. Y is preferably selected from (meth)acrylate and alkyl (meth)acrylates wherein the alkyl groups have from 1 to 8 carbon atoms, including, but not limited to, n-butyl (meth)acrylate, isobutyl (meth)acrylate, methyl (meth)acrylate, and ethyl (meth)acrylate. (113) Preferred O/X and O/X/Y-type copolymers include, without limitation, ethylene acid copolymers, such as ethylene/(meth)acrylic acid, ethylene/(meth)acrylic acid/maleic anhydride, ethylene/(meth)acrylic acid/maleic acid mono-ester, ethylene/maleic acid, ethylene/maleic acid mono-ester, ethylene/(meth)acrylic acid/n-butyl (meth)acrylate, ethylene/(meth)acrylic acid/isobutyl (meth)acrylate, ethylene/(meth)acrylic acid/methyl (meth)acrylate, ethylene/(meth)acrylic acid/ethyl (meth)acrylate terpolymers, and the like. The term, "copolymer," as used herein, includes polymers having two types of monomers, those having three types of monomers, and those having more than three types of monomers. Preferred α , β -ethylenically unsaturated mono- or dicarboxylic acids are (meth)acrylic acid, ethacrylic acid, maleic acid, crotonic acid, fumaric acid, itaconic acid. (Meth)acrylic acid is most preferred. As used herein, "(meth)acrylic acid" means methacrylic acid and/or acrylic acid. Likewise, "(meth)acrylate" means methacrylate and/or acrylate.

(114) In a particularly preferred version, highly neutralized E/X- and E/X/Y-type acid copolymers, wherein E is ethylene, X is a C.sub.3-C.sub.8 α , β -ethylenically unsaturated carboxylic acid, and Y is a softening monomer are used. X is preferably selected from methacrylic acid, acrylic acid, ethacrylic acid, crotonic acid, and itaconic acid. Methacrylic acid and acrylic acid are particularly preferred. Y is preferably an acrylate selected from alkyl acrylates and aryl acrylates and preferably selected from (meth)acrylate and alkyl (meth)acrylates wherein the alkyl groups have from 1 to 8 carbon atoms, including, but not limited to, n-butyl (meth)acrylate, isobutyl (meth)acrylate, methyl (meth)acrylate, and ethyl (meth)acrylate. Preferred E/X/Y-type copolymers are those wherein X is (meth)acrylic acid and/or Y is selected from (meth)acrylate, n-butyl (meth)acrylate, isobutyl (meth)acrylate, methyl (meth)acrylate, and ethyl (meth)acrylate. More preferred E/X/Y-type copolymers are ethylene/(meth)acrylic acid/n-butyl acrylate, ethylene/(meth)acrylic acid/methyl acrylate, and ethylene/(meth)acrylic acid/ethyl acrylate.

(115) The amount of ethylene in the acid copolymer may be at least about 15 weight percent, at least about 25 weight percent, at least about 40 weight percent, or at least about 60 weight percent, based on total weight of the copolymer. The amount of C.sub.3 to C.sub.8 α , β -ethylenically unsaturated mono- or dicarboxylic acid in the acid copolymer is typically from 1 weight percent to 35 weight percent, from 5 weight percent to 30 weight percent, from 5 weight percent to 25 weight percent, or from 10 weight percent to 20 weight percent, based on total weight of the copolymer. The amount of optional softening comonomer in the acid copolymer may be from 0 weight percent to 50 weight percent, from 5 weight percent to 40 weight percent, from 10 weight percent to 35 weight percent, or from 20 weight percent to 30 weight percent, based on total weight of the copolymer.

(116) The various O/X, E/X, O/X/Y, and E/X/Y-type copolymers are at least partially neutralized with a cation source, optionally in the presence of a high molecular weight organic acid, such as those disclosed in U.S. Pat. No. 6,756,436, the entire disclosure of which is hereby incorporated herein by reference. The acid copolymer can be reacted with the optional high molecular weight organic acid and the cation source simultaneously, or prior to the addition of the cation source. Suitable cation sources include, but are not limited to, metal ion sources, such as compounds of alkali metals, alkaline earth metals, transition metals, and rare earth elements; ammonium salts and monoamine salts; and combinations thereof. Preferred cation sources are compounds of magnesium, sodium, potassium, cesium, calcium, barium, manganese, copper, zinc, lead, tin, aluminum, nickel, chromium, lithium, and rare earth metals. The amount of cation used in the composition is readily determined based on desired level of neutralization. As discussed above, for HNP compositions, the acid groups are neutralized to 70 percent or greater, 70 to 100 percent, or 90

to 100 percent. In one embodiment, an excess amount of neutralizing agent, that is, an amount greater than the stoichiometric amount needed to neutralize the acid groups, may be used. That is, the acid groups may be neutralized to 100 percent or greater, for example 110 percent or 120 percent or greater. In other embodiments, partially neutralized compositions are prepared, wherein 10 percent or greater, normally 30 percent or greater of the acid groups are neutralized. When aluminum is used as the cation source, it is preferably used at low levels with another cation such as zinc, sodium, or lithium, since aluminum has a dramatic effect on melt flow reduction and cannot be used alone at high levels. For example, aluminum is used to neutralize about 10 percent of the acid groups and sodium is added to neutralize an additional 90 percent of the acid groups.

(117) “Low acid” and “high acid” ionomeric polymers, as well as blends of such ionomers, may be used. In general, low acid ionomers are considered to be those containing 16 weight percent or less of acid moieties, whereas high acid ionomers are considered to be those containing greater than 16 weight percent of acid moieties. In one embodiment, the inner cover layer is formed from a composition comprising a high acid ionomer. A suitable high acid ionomer is Surlyn® 8150.

(Dow), which is a copolymer of ethylene and methacrylic acid, having an acid content of 19 weight percent, 45 percent neutralized with sodium. In another embodiment, the inner cover layer is formed from a composition comprising a high acid ionomer and a maleic anhydride-grafted non-ionomeric polymer. An example of a suitable maleic anhydride-grafted polymer is Fusabond® 525D (Dow), which is a maleic anhydride-grafted, metallocene-catalyzed ethylene-butene copolymer having about 0.9 weight percent maleic anhydride grafted onto the copolymer. Blends of high acid ionomers with maleic anhydride-grafted polymers are further disclosed, for example, in U.S. Pat. Nos. 6,992,135 and 6,677,401, the entire disclosures of which are hereby incorporated herein by reference.

(118) The inner cover layer also may be formed from a composition comprising a 50/45/5 blend of Surlyn® 8940/Surlyn® 9650/Nucrel® 960. In this aspect, the composition may have a material hardness of from 80 to 85 Shore C. In another embodiment, the inner cover layer is formed from a composition comprising a 50/25/25 blend of Surlyn® 8940/Surlyn® 9650/Surlyn® 9910, having a material hardness of about 85 to 95 Shore C. In yet another embodiment, the inner cover layer is formed from a composition comprising a 50/50 blend of Surlyn® 8940/Surlyn® 9650, having a material hardness of about 82 to 90 Shore C. A composition comprising a 50/50 blend of Surlyn® 8940 and Surlyn® 7940 also may be used.

(119) The compositions used to make the layers outside of the core, e.g., the outer cover layer and, when present, the inner cover layer, may contain a variety of fillers and additives to impart specific properties to the ball. For example, relatively heavy-weight and light-weight metal fillers such as, particulate; powders; flakes; and fibers of copper, steel, brass, tungsten, titanium, aluminum, magnesium, molybdenum, cobalt, nickel, iron, lead, tin, zinc, barium, bismuth, bronze, silver, gold, and platinum, and alloys and combinations thereof may be used to adjust the specific gravity of the ball. Other additives and fillers include, but are not limited to, optical brighteners, coloring agents, fluorescent agents, whitening agents, UV absorbers, light stabilizers, surfactants, processing aids, antioxidants, stabilizers, softening agents, fragrance components, plasticizers, impact modifiers, titanium dioxide, clay, mica, talc, glass flakes, milled glass, and mixtures thereof.

(120) The outer cover layer preferably has a material hardness of 85 Shore C or less. The thickness of the outer cover layer is preferably within a range having a lower limit of 0.010 or 0.015 or 0.025 inches and an upper limit of 0.035 or 0.040 or 0.055 or 0.080 inches. Methods for measuring hardness of the layers in the golf ball are described in further detail above. When included, the inner cover layer preferably has a material hardness within a range having a lower limit of 70 or 75 or 80 or 82 Shore C and an upper limit of 85 or 86 or 90 or 92 Shore C. The thickness of the intermediate layer is preferably within a range having a lower limit of 0.010 or 0.015 or 0.020 or 0.030 inches and an upper limit of 0.035 or 0.045 or 0.080 or 0.120 inches.

(121) In one embodiment, the golf balls made in accordance with the present disclosure include a

core comprising a rubber formulation as described herein, an inner cover layer formed from an ionomeric material, and the outer cover layer is formed from a polyurethane material, and the outer cover layer has a hardness that is less than that of the inner cover layer. For example, the inner cover layer may have a hardness of greater than about 60 Shore D and the outer cover layer may have a hardness of less than about 60 Shore D. In an alternative embodiment, the inner cover layer is comprised of a partially or fully neutralized ionomer, a thermoplastic polyester elastomer, a thermoplastic polyether block amide, or a thermoplastic or thermosetting polyurethane or polyurea, and the outer cover layer is comprised of an ionomeric material. In this alternative embodiment, the inner cover layer may have a hardness of less than about 60 Shore D and the outer cover layer may have a hardness of greater than about 55 Shore D and the inner cover layer hardness is less than the outer cover layer hardness.

(122) When a dual cover is disposed about the core, the inner cover layer may have a thickness of about 0.01 inches to about 0.06 inches, about 0.015 inches to about 0.040 inches, or about 0.02 inches to about 0.035 inches. The outer cover layer may have a thickness of about 0.015 inches to about 0.055 inches, about 0.02 inches to about 0.04 inches, or about 0.025 inches to about 0.035 inches.

(123) The golf balls of the present disclosure may be formed using a variety of application techniques. For example, the golf ball, golf ball core, or any layer of the golf ball may be formed using compression molding, flip molding, injection molding, retractable pin injection molding, reaction injection molding (RIM), liquid injection molding (LIM), casting, vacuum forming, powder coating, flow coating, spin coating, dipping, spraying, and the like. Conventionally, compression molding and injection molding are applied to thermoplastic materials, whereas RIM, liquid injection molding, and casting are employed on thermoset materials. In this aspect, cover layers may be formed over the core using any suitable technique that is associated with the material used to form the layer. Preferably, each cover layer is separately formed over the core. For example, an ethylene acid copolymer ionomer composition may be injection-molded to produce half-shells over the core. Alternatively, the ionomer composition can be placed into a compression mold and molded under sufficient pressure, temperature, and time to produce the hemispherical shells, which may then be placed around the core in a compression mold. An outer cover layer including a polyurethane or polyurea composition over the ball sub-assembly may be formed by using a casting process.

(124) Golf balls made in accordance with the present disclosure may be subjected to finishing steps such as flash-trimming, surface-treatment, marking, coating, and the like using techniques known in the art. In one embodiment, a white-pigmented cover may be surface-treated using a suitable method such as, for example, corona, plasma, or ultraviolet (UV) light-treatment. Indicia such as trademarks, symbols, logos, letters, and the like may be printed on the cover using pad-printing, ink-jet printing, dye-sublimation, or other suitable printing methods. Clear surface coatings (for example, primer and topcoats), which may contain a fluorescent whitening agent, may be applied to the cover. Golf balls may also be painted with one or more paint coatings in a variety of colors. In one embodiment, white primer paint is applied first to the surface of the ball and then a white top-coat of paint may be applied over the primer.

EXAMPLES

(125) The invention is further illustrated by the following examples. It should be understood that the examples below are for illustrative purposes only. These examples should not be construed as limiting the scope of the invention.

(126) The golf ball cores of these examples are made using rubber formulations in accordance with the present disclosure. The compositions of the rubber formulations of each golf ball core included in the examples below is provided. These examples provide the components included in each rubber formulation. Concentrations of each component are provided in parts by weight per 100 parts of the total rubber unless stated otherwise.

Example 1

(127) Table 1 provides examples of compositions of golf ball cores having a single layer (solid sphere) or the inner core of a dual-layer core made from rubber formulations in accordance with the present disclosure. The rubber formulations in Table 1 below include a base rubber comprised of polybutadiene rubber and in some embodiments, a co-agent, zinc diacrylate; a complexing agent; a filler; a radical scavenger, Zn-PCTP-72; an initiator, dicumyl peroxide; a water releasing agent; and a hardening agent. The amounts in are in phr.

(128) TABLE-US-00001 TABLE 1 Cores Including Water Releasing Agent and Hardening Agent A B C D Composition Rubber 100 95 100 95 Complexing Agent 5 5 5 5 Filler 12.1 12.1 12.1 12.1 ZnPCTP 0.35 0.35 0.35 0.35 Zinc Diacrylate 32 32 32 32 Dicumyl Peroxide 0.8 0.8 0.8 0.8 Water Releasing Agent 3.0 3.0 1.0 3.0 Hardening Agent 3.0 1.0 1.0 3.0

(129) As shown in Table 1, the ratio of water releasing agent to hardening agent may differ. Formulation D may also be used as a dual core with an outer core layer formed using the formulation E in Table 2 below. The amounts are in phr.

(130) TABLE-US-00002 TABLE 2 Outer Core Formulation E Polybutadiene Rubber 81 Styrene-Butadiene Rubber 19 Filler 14.8 ZnPCTP 0.71 Zinc Diacrylate 35.5 Dicumyl Peroxide 0.35 Regrind 28

(131) Notwithstanding the above, a person having ordinary skill in the art would understand that the relative amount of each component used would depend on the desired properties of the resulting golf ball core and the desired performance characteristics of the golf ball made therefrom.

Example 2

(132) Table 4 below provides various measured properties of golf ball cores made from the formulations in Table 3 in accordance with the present disclosure. The component amounts are presented in phr.

(133) TABLE-US-00003 TABLE 3 Core Formulations Control 1 2 3 CB 1221 100 100 100 100 Perkadox BC 0.80 0.80 0.80 0.80 ZnPCTP 0.35 0.35 0.35 0.35 ZDA 32 32 32 32 ZnO 5 5 5 Abietic Acid — 1 1 3 Calcium Sulfate — 1 3 3 Barium Sulfate 13.93 13.31 11.92 12.05

(134) TABLE-US-00004 TABLE 4 Properties of Cores formed from Compositions of Table 3. Gradient Outer Center (Outer – Compression Hardness Hardness Center Core Formulation (DCM) (Shore C) COR (Shore C) Hardness) Comp. Ex A Control 84.4 90.7 0.809 59.9 30.8 Ex. A 1 68.1 87.7 0.787 51.8 35.9 Ex. B 2 56.3 86.7 0.78 49.8 36.9 Ex. C 3 41.4 81.5 0.767 46.6 34.8

FIGS. 5-7 graphically illustrate the relationship between compression and hardness gradient, COR and compression, and amount of hardening agent and hardness gradient. More specifically, as shown in FIGS. 5 and 7, a core formed from a composition including a 1:3 ratio of hardening agent to water releasing agent appears to produce the largest core hardness gradient (compared to the control and other example cores) and a sizeable drop in compression (compared to the control) whereas a core formed from a composition including a 1:1 ratio (1 phr hardening agent to 1 phr water releasing agent) still produces both an increase in the hardness gradient and a considerable decrease in compression (over that of the control). FIG. 6 indicates that more hardening agent reduces compression and COR, regardless of the amount of water releasing agent in the formulation. FIG. 7 shows that a larger amount of hardening agent produces the least increase in the hardness gradient (as compared to the control) of the three inventive formulations.

(135) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art of this disclosure. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the specification and should not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Well known functions or constructions may not be described in detail for brevity or clarity.

(136) The terms “about” and “approximately” shall generally mean an acceptable degree of error or

variation for the quantity measured given the nature or precision of the measurements. Numerical quantities given in this description are approximate unless stated otherwise, meaning that the term “about” or “approximately” can be inferred when not expressly stated.

(137) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well (i.e., at least one of whatever the article modifies), unless the context clearly indicates otherwise.

(138) The terms “first,” “second,” and the like are used to describe various features or elements, but these features or elements should not be limited by these terms. These terms are only used to distinguish one feature or element from another feature or element. Thus, a first feature or element discussed below could be termed a second feature or element, and similarly, a second feature or element discussed below could be termed a first feature or element without departing from the teachings of the disclosure. Likewise, terms like “top” and “bottom”; “front” and “back”; and “left” and “right” are used to distinguish certain features or elements from each other, but it is expressly contemplated that a top could be a bottom, and vice versa.

(139) The golf balls described and claimed herein are not to be limited in scope by the specific embodiments herein disclosed, since these embodiments are intended as illustrations of several aspects of the disclosure. Any equivalent embodiments are intended to be within the scope of this disclosure. Indeed, various modifications of the device in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the appended claims. All patents and patent applications cited in the foregoing text are expressly incorporated herein by reference in their entirety. Any section headings herein are provided only for consistency with the suggestions of 37 C.F.R. § 1.77 or otherwise to provide organizational queues. These headings shall not limit or characterize the invention(s) set forth herein.

Claims

1. A golf ball, comprising: a core formed from a composition comprising: a base rubber; a hardening agent, wherein the hardening agent comprises tricyclic compound and a carboxylic acid, and wherein the tricyclic compound comprises at least one carbon-carbon double bond; and a water releasing agent, and a cover layer disposed about the rubber layer.
2. The golf ball of claim 1, wherein the hardening agent is present in an amount of about 0.5 and about 5 parts per hundred rubber.
3. The golf ball of claim 1, wherein the tricyclic compound comprises a conjugated diene.
4. The golf ball of claim 3, wherein the hardening agent comprises abietic acid, palustric acid, levopimaric acid, or combinations thereof.
5. The golf ball of claim 1, wherein the water releasing agent comprises a metal salt hydrate.
6. The golf ball of claim 1, wherein the water releasing agent is present in an amount of about 0.5 and about 5.0 parts per hundred rubber.
7. The golf ball of claim 1, wherein the composition further comprises a complexing agent.
8. The golf ball of claim 7, wherein the complexing agent comprises a metal oxide.
9. The golf ball of claim 8, wherein the metal oxide is zinc oxide.
10. The golf ball of claim 9, wherein the core comprises a zinc rosinate complex.
11. The golf ball of claim 1, wherein the core further comprises a geometric center having a hardness H.sub.C, a geometric surface having a hardness H.sub.S, and a hardness gradient H.sub.Gr equal to the difference in the geometric center hardness and the surface hardness, wherein the hardness gradient is between about 20 Shore C and about 40 Shore C.
12. The golf ball of claim 11, wherein H.sub.C is in the range of about 35 Shore C to about 65 Shore C, H.sub.S is in the range of about 60 Shore C to about 90 Shore C, and H.sub.S is greater

than H.sub.C.

13. A golf ball, comprising: a core formed from a rubber formulation comprising: a base rubber; a hardening agent comprising a tricyclic compound and a carboxylic acid, wherein the tricyclic compound comprises a conjugated diene; a water releasing agent; and a complexing agent, wherein the complexing agent comprises a metal oxide, wherein the core has a geometric center hardness H.sub.C, a surface hardness H.sub.S, and a hardness gradient H.sub.Gr equal to the difference between H.sub.C and H.sub.S and between about 20 and 45, and wherein the core comprises a metal rosinate complex; and a cover layer disposed about the core.

14. The golf ball of claim 13, wherein the hardening agent comprises abietic acid, palustric acid, levopimaric acid, or combinations thereof.

15. The golf ball of claim 13, wherein the water releasing agent is metal salt hydrate.

16. The golf ball of claim 15, wherein the metal salt hydrate is present in an amount of about 0.5 and about 5.0 parts per hundred rubber and the hardening agent is present in an amount of about 0.5 to about 5 parts per hundred rubber.

17. A golf ball, comprising: a core formed from a rubber formulation comprising: a base rubber; a hardening agent comprising a 20-carbon tricyclic carboxylic acid, isomers and diastereomers thereof, or combinations thereof; and a water releasing agent, wherein the hardening agent and water releasing agent are present in the rubber formulation in a ratio of about 1:1 to about 1:5, wherein the core has a geometric center hardness H.sub.C, a surface hardness H.sub.S, and a hardness gradient H.sub.Gr equal to the difference between H.sub.C and H.sub.S and between about 10 and 50; and a cover layer disposed about the core.

18. The golf ball of claim 17, wherein the formulation further comprises a complexing agent.

19. The golf ball of claim 18, wherein the complexing agent comprises a metal oxide.

20. The golf ball of claim 19, wherein the core comprises a metal rosinate complex of Formula I:

$$HA \cdot X \cdot HA \quad (I)$$
 where HA is a deprotonated form of the hardening agent and X is the metal in the complexing agent.

21. The golf ball of claim 20, wherein X comprises zinc, iron, aluminum, titanium, magnesium, or zirconium.
